

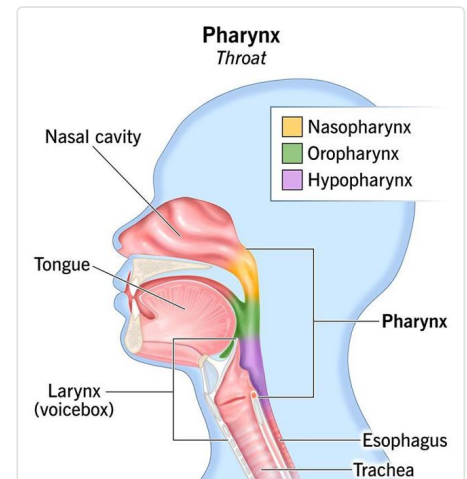
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Name the four tonsils of Waldeyer's ring and where each sits.
2. At how many cm from the incisors does an endoscope meet each esophageal narrowing?
3. Which two veins make the porto-systemic anastomosis at the lower esophagus?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Pharynx — the big picture

What	Funnel-shaped fibromuscular tube : base of skull → esophagus at the lower border of C6
Role	Common channel for food (deglutition) and air (respiration)
Size	12–14 cm long · 3.5 cm wide at its base → 1.5 cm at the pharyngo-esophageal junction
Parts (top → bottom)	Nasopharynx (behind nose) · Oropharynx (behind mouth) · Laryngopharynx (behind larynx)

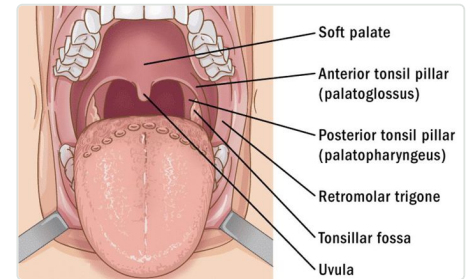


The three parts of the pharynx.

2 The three parts compared

	Nasopharynx	Oropharynx	Laryngopharynx
Extent	Behind the nasal cavities, above the soft palate	Under surface of soft palate → upper border of epiglottis	Upper border of epiglottis → lower border of cricoid (front) / C6 (back)
Function	Respiratory only; resembles the nose	Common path for food and air	Opens into larynx (laryngeal inlet) and esophagus
Wall / lining	Pseudostratified ciliated columnar ; rigid, non-collapsible walls keep the airway open	Walls below	
Roof / floor	Wall boundaries are listed as headings only on the slides — see the lecture	Roof : soft palate (under surface) + pharyngeal isthmus (to nasopharynx) Floor : posterior ⅓ of tongue + interval between tongue and epiglottis	—
Anterior		Incomplete: oropharyngeal isthmus (to mouth) + pharyngeal part of tongue	Laryngeal inlet + posterior surface of larynx
Posterior · lateral		Post: C2 body + upper C3 Lat: pterygomandibular raphe, mandible, tongue, hyoid; palatine tonsil	Post: C3–C6 bodies Lat: thyroid cartilage + thyrohyoid membrane; piriform fossa

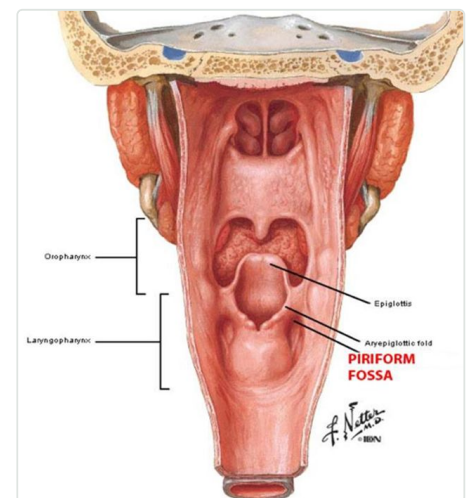
Tonsillar fossa	Triangular fossa on the lateral oropharyngeal wall holding the palatine tonsil : palatoglossal arch in front, palatopharyngeal arch behind
Oropharyngeal isthmus	Closed during swallowing → food cannot regurgitate into the mouth
Crossing paths	Food and air cross in the oropharynx → food may enter the airway (choking); air may enter the gut (gas in the stomach)
PE junction	Pharyngo-esophageal junction = narrowest part of the GIT except the appendix



Oral view: palatoglossal (anterior) and palatopharyngeal (posterior) pillars with the tonsillar fossa between them.

3 Piriform fossa

What	Deep recess in the lateral wall of the laryngopharynx on each side of the laryngeal inlet, made by the larynx bulging into the laryngopharynx
Medial	Aryepiglottic fold + quadrangular membrane
Lateral	Mucosa over the thyroid lamina and thyrohyoid membrane
Above	Separated from the epiglottic vallecula by the lateral glossoepiglottic fold
Under its floor	Internal laryngeal nerve + superior laryngeal vessels (pierce the thyrohyoid membrane)
Clinical	Dangerous site for endoscope perforation · fish bones and pins lodge here · careless removal injures the internal laryngeal nerve → supraglottic anaesthesia + loss of the cough reflex



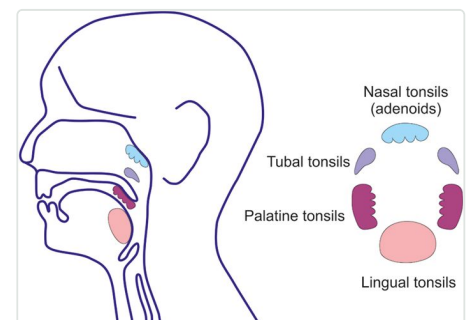
Posterior view of the opened pharynx showing the piriform fossa.

4 Waldeyer's ring

An **interrupted circle of lymphoid tissue** under the pharyngeal lining, guarding the start of the air and food passages against microorganisms.

Position	Tonsil
Postero-superior	Pharyngeal tonsil (nasopharyngeal tonsil / adenoid) — in the roof of the nasopharynx
Anterior	Lingual tonsil
Lateral	Tubal and palatine tonsils

Lecture case: a child with mouth breathing, snoring and nasal obstruction from hypertrophied lymphoid tissue in the nasopharyngeal roof = enlarged **pharyngeal tonsil**.



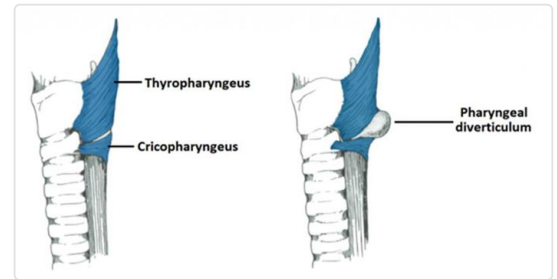
The tonsils making up Waldeyer's ring.

5 Pharyngeal constrictors

Muscle	Origin	Action · clinical
Superior	Pterygoid hamulus, mylohyoid line of mandible, side of tongue	Constricts upper pharynx · weakness → nasopharyngeal regurgitation
Middle	Greater and lesser horns of hyoid, stylohyoid ligament	Constricts middle pharynx during swallowing
Inferior	Thyropharyngeus : oblique line of thyroid cartilage Cricopharyngeus : cricoid cartilage	Constricts lower pharynx · cricopharyngeus = upper esophageal sphincter (keeps air out of the esophagus)

All three: **insert into the pharyngeal raphe** and are supplied by the **pharyngeal plexus via the vagus (CN X)**.

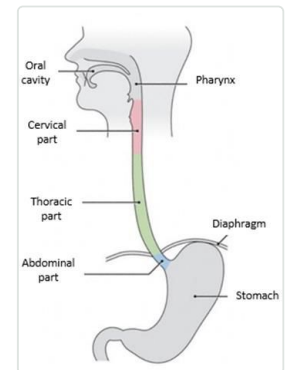
Killian's dehiscence — triangular weak area in the **posterior** pharyngeal wall **between thyropharyngeus and cricopharyngeus**. Mucosa herniates backwards here → **pharyngeal pouch (diverticulum)**: food collects → **dysphagia**.



Thyropharyngeus and cricopharyngeus, and a pharyngeal diverticulum between them.

6 Esophagus — course and parts

What	Muscular tube, 10 in (25 cm) : laryngopharynx → stomach
Course	Begins at the cricoid (C6) → neck → superior mediastinum → posterior mediastinum → pierces the diaphragm at T10 → joins the stomach at the cardiac orifice (T11)



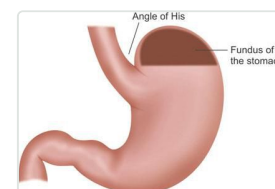
Cervical, thoracic and abdominal parts.

	Cervical (4 cm)	Thoracic (20 cm, longest)	Abdominal (1.5-3 cm, shortest)
Extent	Hypopharynx → sternal notch	Sternal notch (T1) → T10	In front of left lobe of liver and left crus
Anterior	Trachea + recurrent laryngeal nerve	Trachea + left recurrent laryngeal nerve	—
Posterior	C6 and C7 vertebrae	Thoracic vertebral bodies, thoracic duct , azygos veins, descending aorta	
Sides	Thyroid lobes, carotid sheath	Right: mediastinal pleura · Left: left subclavian artery, aortic arch, thoracic duct	
Arteries	Inferior thyroid	Esophageal branches of the descending thoracic aorta	Left gastric (celiac) + left inferior phrenic
Veins	Inferior thyroid	Azygos veins	Left gastric (portal) + azygos (systemic)
Lymph	Deep cervical nodes	Superior and posterior mediastinal nodes	Posterior mediastinal + left gastric nodes
Nerves	Recurrent laryngeal nerves	Parasympathetic and sympathetic fibres via the vagi	Sympathetic: greater splanchnic (T5-T9) · parasympathetic: vagal trunks

Porto-systemic anastomosis at the lower esophagus. Submucosal veins drain both to the **portal** system (left gastric vein) and the **systemic** system (azygos vein). When portal pressure rises they dilate into **esophageal varices**, which can rupture → **haematemesis**.

7 Sphincters and the anti-reflux barrier

	Upper esophageal sphincter	Lower esophageal sphincter
Type	Anatomical, striated	Physiological
Formed by	Cricopharyngeus	Gastro-esophageal junction (T11): the principal internal barrier
Helped by	—	Right crus of the diaphragm + angle of His



Angle of His.

Crural diaphragm	External sphincter: the right crus forms the main muscle around the esophageal hiatus (T10)
Angle of His	Acute angle between the left side of the abdominal esophagus and the fundus → flap valve

GERD (pyrosis)	Low LES pressure or inappropriate transient LES relaxations → gastric contents reflux → inflammation and ulcers
Hiatal hernia	Part of the stomach and/or GE junction slides through the hiatus into the thorax
Achalasia of the cardia	The GE sphincter fails to relax → obstruction to flow into the stomach

8 Esophageal constrictions and endoscopy

DISTANCE LADDER — each narrowing, in cm from the incisor teeth

0 cm		incisor teeth
15 cm	C6	cricopharyngeus = UES
22.5 cm	T4	arch of the aorta
27.5 cm	T5	left main bronchus
40 cm	T10/11*	diaphragmatic hiatus → GE junction

*Slides: diaphragmatic constriction at T11 in the text, T10 in the figure.
The esophagus pierces the diaphragm at T10 and joins the stomach at T11.

Esophagus = 25 cm

cervical — 4 cm
thoracic — 20 cm (longest)
abdominal — 1.5–3 cm (shortest)

Mnemonic ABCD

Aortic arch · Bronchus (left)
Cricopharyngeus · Diaphragm

Constriction	Cause	Level	From incisors
Cervical	Cricopharyngeus (UES) at the PE junction	C6	15 cm
Thoracic (broncho-aortic)	Compound: arch of aorta , then left main bronchus	T4 · T5	22.5 cm · 27.5 cm
Diaphragmatic	Esophageal hiatus	T11 (text) / T10 (figure)	40 cm

EGD (esophago-gastro-duodenoscopy) reaches the **start of the 3rd part of the duodenum**; allows biopsy and diagnosis of varices. The esophagus is 25 cm long, but the GE junction lies **40 cm from the incisors**.

P Must-remember pearls

- Pharynx: base of skull → **lower border of C6**; 12–14 cm; PE junction is the **narrowest part of the GIT except the appendix**.
- Palatine tonsil sits between the **palatoglossal (front)** and **palatopharyngeal (back)** arches on the lateral oropharyngeal wall.
- Piriform fossa: internal laryngeal nerve under its floor → injury = **loss of the cough reflex**.

- Constrictors insert into the **pharyngeal raphe** and are supplied by the **pharyngeal plexus (CN X)**; **cricopharyngeus = UES**.
- **Killian's dehiscence**: between thyropharyngeus and cricopharyngeus → pharyngeal pouch → dysphagia.
- Esophagus 25 cm: **C6 → pierces diaphragm T10 → cardiac orifice T11**; cervical 4 cm, thoracic 20 cm, abdominal 1.5–3 cm.
- Cervical esophagus → **deep cervical nodes** · varices = **left gastric vein (portal) ↔ azygos (systemic)**.
- UES = anatomical (cricopharyngeus) · LES = physiological, helped by the **right crus** and the **angle of His**.
- Distances: **15 · 22.5 · 27.5 · 40 cm** = cricopharyngeus · aortic arch · left bronchus · diaphragm (ABCD).

Q Test yourself

1. A 5-year-old boy has chronic mouth breathing, loud snoring and nasal obstruction. The lymphoid tissue in the roof of his nasopharynx is hypertrophied. Which tonsil is enlarged?

- | | |
|---------------------|----------------------|
| A. Palatine tonsil | B. Lingual tonsil |
| C. Tubal tonsil | D. Pharyngeal tonsil |
| E. Laryngeal tonsil | |

2. After a fish bone is removed from a patient's piriform fossa, he coughs poorly when fluid touches his supraglottic larynx. Which nerve was injured?

- | | |
|------------------------------|-----------------------------|
| A. Recurrent laryngeal nerve | B. Internal laryngeal nerve |
| C. External laryngeal nerve | D. Hypoglossal nerve |
| E. Glossopharyngeal nerve | |

3. A 70-year-old man has dysphagia and regurgitates food he ate days earlier. A barium swallow shows a pouch arising from the posterior pharyngeal wall. Between which two muscles did it herniate?

- | | |
|--|---|
| A. Superior and middle constrictors | B. Middle constrictor and stylopharyngeus |
| C. Thyropharyngeus and cricopharyngeus | D. Cricopharyngeus and esophageal circular muscle |
| E. Palatoglossus and palatopharyngeus | |

4. During endoscopy, a smooth extrinsic narrowing is seen 27.5 cm from the incisor teeth. What structure causes it?

- | | |
|-------------------------|------------------------|
| A. Arch of the aorta | B. Cricopharyngeus |
| C. Diaphragmatic hiatus | D. Right main bronchus |
| E. Left main bronchus | |

5. A 52-year-old man with cirrhosis vomits a large amount of blood from dilated veins in his lower esophagus. Which vein carries this blood into the portal system?

- | | |
|-----------------------|--------------------------|
| A. Left gastric vein | B. Azygos vein |
| C. Hemiazygos vein | D. Inferior thyroid vein |
| E. Superior vena cava | |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 D — The pharyngeal tonsil (adenoid) lies postero-superiorly in the nasopharyngeal roof.

2 B — The internal laryngeal nerve runs under the floor of the piriform fossa; damage gives supraglottic anaesthesia and a lost cough reflex.

3 C — Killian's dehiscence, a weak area between thyropharyngeus and cricopharyngeus, is where a pharyngeal pouch forms.

4 E — The thoracic constriction is compound: aortic arch at 22.5 cm (T4), then the left main bronchus at 27.5 cm (T5).

5 A — The left gastric vein is the portal side of the lower-esophageal anastomosis; the azygos is the systemic side.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam